

28 Supersymmetric Versions of the Standard Model

28.4 The Minimal Supersymmetric Standard Model

In the previous section we identified two different ways that the breakdown of supersymmetry at a high energy scale M_S could be communicated to the known quarks and leptons: through gauge or gravitational superfields. The supersymmetry-breaking terms in the resulting low-energy effective Lagrangian will then be suppressed by powers of gauge couplings or the Newton constant. Most of these terms will therefore be rather small, with the exception that, along with factors of gauge couplings or the Newton constant, on dimensional grounds the mass terms and other superrenormalizable terms in the effective Lagrangian will be proportional to one or more factors of the supersymmetry-breaking scale M_S , which is quite large compared with known particle masses. We can conclude then that, to a fair approximation for gauge-mediated supersymmetry breaking and a very good approximation for gravitationally mediated supersymmetry breaking, the main effect of supersymmetry breaking will be in the superrenormalizable terms of the effective Lagrangian of the supersymmetric standard model. This version of the standard model [10], which is supersymmetric except for superrenormalizable terms, is usually known as the *minimal supersymmetric standard model*.

With R parity or $B - L$ conserved, the most general superrenormalizable Lagrangian density allowed by $SU(3) \cdot SU(2) \cdot U(1)$ gauge symmetry takes the form

$$\begin{aligned}
\mathcal{L}_{\text{SR}} = & - \sum_{ij} M_{ij}^{2Q} (\mathcal{Q}_i^\dagger \mathcal{Q}_j) - \sum_{ij} M_{ij}^{2\bar{U}} (\bar{\mathcal{U}}_i^\dagger \bar{\mathcal{U}}_j) - \sum_{ij} M_{ij}^{2\bar{D}} (\bar{\mathcal{D}}_i^\dagger \bar{\mathcal{D}}_j) \\
& - \sum_{ij} M_{ij}^{2L} (\mathcal{L}_i^\dagger \mathcal{L}_j) - \sum_{ij} M_{ij}^{2\bar{E}} (\bar{\mathcal{E}}_i^\dagger \bar{\mathcal{E}}_j) \\
& - (\bar{\lambda}_3 m_{\text{gluino}} \lambda_3) - (\bar{\lambda}_2 m_{\text{wino}} \lambda_2) - (\bar{\lambda}_1 m_{\text{bino}} \lambda_1) \\
& - \sum_{ij} A_{ij}^D h_{ij}^D (\mathcal{Q}_i^T e \mathcal{H}_1) \bar{\mathcal{D}}_j - \sum_{ij} A_{ij}^E h_{ij}^E (\mathcal{L}_i^T e \mathcal{H}_1) \bar{\mathcal{E}}_j \\
& - \sum_{ij} A_{ij}^U h_{ij}^U (\mathcal{Q}_i^T e \mathcal{H}_2) \bar{\mathcal{U}}_j - \sum_{ij} C_{ij}^D h_{ij}^D (\mathcal{Q}_i^T \mathcal{H}_2^*) \bar{\mathcal{D}}_j \\
& - \sum_{ij} C_{ij}^E h_{ij}^E (\mathcal{L}_i^T \mathcal{H}_2^*) \bar{\mathcal{E}}_j - \sum_{ij} C_{ij}^U h_{ij}^U (\mathcal{Q}_i^T \mathcal{H}_1^*) \bar{\mathcal{U}}_j \\
& - B\mu (\mathcal{H}_2^T e \mathcal{H}_1) + \text{H.c.}
\end{aligned} \tag{28.4.1}$$

Script letters are used here to denote the scalar components of left-chiral superfields. Sums over $SU(2)$ and color indices are understood, as necessary for invariance under $SU(3) \cdot SU(2) \cdot U(1)$, with e the usual antisymmetric $2 \cdot 2$ matrix $i\sigma_2$. All coefficients may be complex, and the gaugino masses may involve terms proportional to the γ_5 as well as the unit matrix.

We follow the custom here of writing the coefficients of the terms involving scalar fields but not their adjoints as equal to the coefficients of the corresponding supersymmetric $\mathcal{F}$ -terms in Eq. (28.1.7) times factors A_{ij}^D , A_{ij}^E , A_{ij}^U , and B . This is motivated by the consideration that the smallness of the Yukawa couplings of the light quarks in Eq. (28.1.7)

reflects a number of approximate chiral symmetries that if extended to the whole supermultiplet would also make the corresponding trilinear terms in Eq. (28.4.1) small, while the appearance of the μ -term in Eq. (28.1.7) violates a possible Peccei–Quinn symmetry [4] that if approximately valid would make both μ and $B\mu$ small. Similar considerations suggest the form in which we write the coefficients of the terms that involve both scalars and their complex conjugates. Also, in Section 31.4 we will describe contributions to Ah and $B\mu$ that really are proportional to h and μ , respectively. However, here we are leaving it an open question whether the Ah , Ch , and $B\mu$ coefficients in Eq. (28.4.1) are necessarily small when the corresponding h and μ coefficients in Eq. (28.1.7) are small.

The Ch -terms in Eq. (28.4.1) have generally been omitted in discussions of the minimal supersymmetric standard model. This is partly because, as discussed in Section 27.7, terms like these that involve the ϕ -components of left-chiral scalar superfields and also their complex conjugates can potentially produce quadratic divergences and thereby raise fine-tuning problems. But we saw in Section 27.7 that the quadratic divergences occur only in ‘tadpole’ graphs in which a scalar field line disappears into the vacuum, and in the minimum supersymmetric standard model there are no scalars that are neutral under all gauge symmetries, and hence no scalar tadpoles. The Ch -terms are absent in the theories of gravitationally mediated supersymmetry breaking discussed in Section 31.6, and small in the theories of gauge-mediated supersymmetry breaking described in Section 28.6, but there is no reason to suppose that this will always be the case.

Even though they are not supersymmetric, superrenormalizable interactions like those in Eq. (28.4.1) are shown in Section 27.7 to not produce supersymmetry-violating ultraviolet-divergent corrections to the coefficients of the supersymmetric $d = 4$ interactions. The condition of supersymmetry that is imposed on the dimensionless couplings of the minimal supersymmetric standard model therefore does not get in the way of the cancellation of ultraviolet divergences by renormalization of coupling constants. It was this property, rather than any theory of supersymmetry breaking in a high energy hidden sector, that motivated the introduction of the minimal supersymmetric standard model in Reference 10.

The best reason today for studying the implications of the supersymmetric standard model is that, as already mentioned, theories in which supersymmetry is spontaneously broken at a high energy scale are naturally described by the minimal supersymmetric standard model at much lower energies. We can explore the phenomenological implications of the supersymmetric standard model, and be reasonably confident that the results will be relevant whatever detailed model of supersymmetry breaking and its mediation turns out to be correct.

Even without the Ch -terms, if all of the other coefficients in the Lagrangian are constrained only by gauge symmetries and R parity conservation, the minimal supersymmetric standard model will contain over 100 free parameters [11]. Here ‘minimal’ means nothing more than that the theory contains only a minimal menu of superfields. Sometimes the term ‘minimal supersymmetric standard model’ is reserved for models that also satisfy restrictions on the coefficients of the superrenormalizable terms, motivated either by some underlying theory or by empirical constraints. For instance, the minimal supersymmetric

standard model is sometimes optimistically assumed to satisfy the universality conditions

$$\begin{aligned}
M_{ij}^{2Q} &= M_{ij}^{2\bar{D}} = M_{ij}^{2\bar{U}} = M_{ij}^{2L} = M_{ij}^{2\bar{E}} = M^2 \delta_{ij}, \\
m_{\text{gluino}} &= m_{\text{wino}} = m_{\text{bino}}, \\
A_{ij}^D &= A_{ij}^E = A_{ij}^U = A, \quad C_{ij}^D = C_{ij}^E = C_{ij}^U = 0.
\end{aligned}
\tag{28.4.2}$$

Often these conditions are imposed at the scale $M_X \approx 10^{16}$ GeV of coupling constant unification, with corrections produced only by the renormalization group flow to lower energies. We will not be making such assumptions here.

In analyzing the phenomenological implications of the minimal supersymmetric standard model, we must deal not only with the search for new particles, but also with two classes of severe empirical constraint on processes involving known particles: the experimental upper bounds on various flavor non-conserving processes, and on various modes of CP non-conservation.

Flavor Changing Processes

We saw in Section 21.3 that there is an automatic suppression of flavor-changing processes like K^0 - $\bar{K}^0$ oscillations and $K^0 \rightarrow \mu^+ \mu^-$ in the non-supersymmetric standard model. This is due to the feature of this theory, that it is only the mass splittings of the quarks that prevent them from being defined so that each flavor is separately conserved, so the amplitude of these flavor-changing process must be proportional to several factors of small quark masses. Also, in this theory lepton flavor is automatically conserved, so that processes like $\mu \rightarrow e \gamma$ are absolutely forbidden. These satisfactory results are put at risk by the presence of squarks and sleptons in supersymmetric extensions of the standard model, because there is in general no reason to expect that the squark and slepton mass matrices will be diagonal in the same basis as the quark and lepton mass matrices. This does not introduce flavor changing in the interaction of these particles with gauge bosons, which are flavor-independent, but it can produce flavor-changing transitions in which squarks or sleptons turn into quarks or leptons with the emission or absorption of gauginos. Of course, there

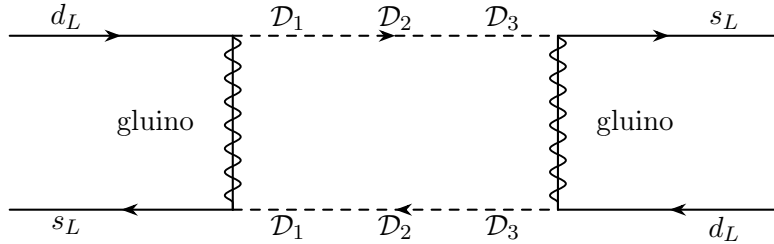


Figure 28.1. A one-loop diagram that can contribute to the $\Delta S = 2$ effective interaction $(\bar{s}_L \gamma^\mu d_L)(\bar{s}_L \gamma_\mu d_L)$ in the supersymmetric standard model. Here solid lines are quarks; dashed lines are squarks; and combined solid and wavy lines are gluinos.

is no problem if the squarks and sleptons are degenerate, in which case their mass matrices are diagonal in any basis.

The most stringent limits on squark mass splittings and/or mixing angles are set by measurements of K^0 - $\bar{K}^0$ transitions [12]. These transitions are produced by operators in the effective low-energy Lagrangian density like $(\bar{s}_L \gamma^\mu d_L)(\bar{s}_L \gamma_\mu d_L)$, which can be produced by diagrams like Figure 28.1. The superpartners of the quarks d_L and s_L are in general linear combinations $\sum_i V_{di} \mathcal{D}_i$ and $\sum_i V_{si} \mathcal{D}_i$ of squarks $\mathcal{D}_i$ of definite mass, where V_{ji} is a $3 \cdot 3$ unitary matrix, so the two squark propagators in this diagram contribute a factor

$$\sum_i \frac{V_{di} V_{si}^*}{k^2 + M_i^2 - i\epsilon} \cdot \sum_j \frac{V_{dj} V_{sj}^*}{k^2 + M_j^2 - i\epsilon},$$

where k is the four-momentum circulating in the loop. Because V_{ji} is unitary, this vanishes if the three squark masses M_i are all equal. If the squark square masses differ from some common value M_{squark}^2 by relatively small amounts ΔM_i^2 , then this becomes

$$\left(\frac{1}{k^2 + M_{\text{squark}}^2 - i\epsilon} \right)^4 \left(\sum_i V_{di} V_{si}^* \Delta M_i^2 \right)^2.$$

The amplitude for $d_L \bar{s}_L \rightarrow s_L \bar{d}_L$ has dimensionality mass^{-2} , so after multiplying by the gluino propagators and four factors of the strong coupling g_s , and integrating over k , we must get an amplitude proportional to

$$\frac{g_s^4}{\widetilde{M}^6} \left(\sum_i V_{di} V_{si}^* \Delta M_i^2 \right)^2, \quad (28.4.3)$$

where $\widetilde{M}$ is the larger of M_{squark} and m_{gluino} . This may be compared with the result for this amplitude in the non-supersymmetric standard model, which is produced by W exchange, as shown in Figure 28.2. Ignoring the third generation of quarks, which has only small transition amplitudes to the first two generations, the amplitudes for $d \rightarrow u$, $d \rightarrow c$, $s \rightarrow u$ and $s \rightarrow c$ by W^- emission are, respectively, $\cos \theta_c$, $-\sin \theta_c$, $\sin \theta_c$, and $\cos \theta_c$, where θ_c is the Cabibbo angle defined in Section 21.3. Hence in place of the squark propagators here we have the quark propagators

$$\sin \theta_c \cos \theta_c \left(\frac{i\not{k} + m_u}{k^2 + m_u^2 - i\epsilon} - \frac{i\not{k} + m_c}{k^2 + m_c^2 - i\epsilon} \right),$$

and in place of the strong coupling g_s here we have the $SU(2)$ coupling g . Thus in the non-supersymmetric standard model the amplitude for $d_L \bar{s}_L \rightarrow s_L \bar{d}_L$ is proportional to

$$\frac{g^4 \sin^2 \theta_c \cos^2 \theta_c}{m_W^4} (m_c - m_u)^2, \quad (28.4.4)$$

with a proportionality coefficient of the same order as that in Eq. (28.4.3). With a plausible guess on how to calculate the K^0 - $\bar{K}^0$ transition amplitude from the amplitude for $d_L \bar{s}_L \rightarrow s_L \bar{d}_L$, the amplitude corresponding to Figure 28.2 is known to give a result in good agreement with experiment. (Indeed, Gaillard and Lee [13] used this calculation to

predict that $m_c \approx 1.5$ GeV, before the c quark was discovered.) It therefore seems reasonable to require that the squark exchange result (28.4.3) should be less than the quark exchange result (28.4.4). This yields the condition

$$\left| \sum_i V_{di} V_{si}^* \frac{\Delta M_i^2}{\widetilde{M}^2} \right| < \frac{g^2}{g_s^2} \sin \theta_c \cos \theta_c \frac{(m_c - m_u) \widetilde{M}}{m_W^2}. \quad (28.4.5)$$

Taking $g^2/4\pi = 0.036$, $g_s^2/4\pi = 0.118$, $\sin \theta_c = 0.22$, $m_W = 80.4$ GeV, $m_c = 1.5$ GeV, and $m_u \ll m_c$, we find

$$\left| \sum_i V_{di} V_{si}^* \frac{\Delta M_i^2}{\widetilde{M}^2} \right| < 1.5 \cdot 10^{-3} \cdot (\widetilde{M}/100 \text{ GeV}). \quad (28.4.6)$$

The squark masses are unlikely to be much less than m_{gluino} , so we can conclude that either the squark masses are split by no more than about one part in 10^3 , or the non-diagonal terms of the mixing matrix V_{ji} are less than about 10^{-3} , or the squarks are heavier than about 10 TeV, or we have some combination of nearly degenerate squarks, nearly zero mixing angles, and heavy squarks. In itself, this result only constrains the superpartners $\mathcal{D}_i$ of the left-handed quarks of charge $-e/3$, but similar limits on the masses and mixing angles of the $\bar{\mathcal{D}}_i$ squarks can be obtained by considering the amplitude for $d_R \bar{s}_R \rightarrow s_R \bar{d}_R$. We can also obtain somewhat weaker

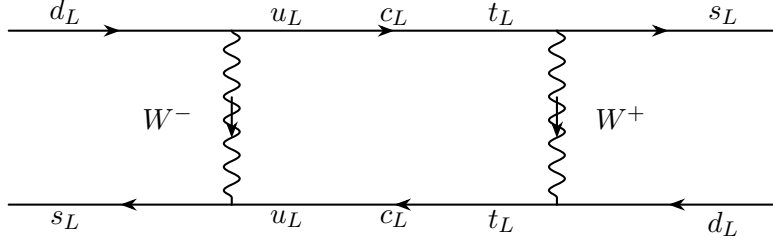


Figure 28.2. A one-loop diagram that can contribute to the $\Delta S = 2$ effective interaction $(\bar{s}_L \gamma^\mu d_L)(\bar{d}_L \gamma_\mu s_L)$ in both the supersymmetric and non-supersymmetric standard models. Here solid lines are quarks and wavy lines are $W^\pm$ bosons.

limits on the masses and mixing angles of the $\mathcal{U}_i$ squarks by considering the amplitudes produced by wino exchange rather than gluino exchange. It should be noted, however, that these arguments put no constraints on the differences between the masses of squarks of different charge, or on the differences between the masses of the superpartners $\mathcal{Q}_i$ and $\bar{\mathcal{Q}}_i$ of left-handed quarks and antiquarks.

Just as for the squarks, the sleptons of definite mass are expected to be non-diagonal linear combinations of the superpartners of the leptons. This leads to the decay process $\mu \rightarrow e + \gamma$ through diagrams like Figure 28.3. The experimental upper bound $4.9 \cdot 10^{-11}$ on the branching ratio of this process then sets a limit of about 10^{-3} on the fractional mass splitting of sleptons of the same charge but different generations for generic mixing angles, or on the mixing angles for non-degenerate sleptons [14].

Attempts have been made to explain the degeneracy of the squarks and sleptons in terms of a gauge symmetry connecting the different generations [14a]. In Section 28.6 we will describe an approach to supersymmetry breaking in which this degeneracy appears without needing to impose such symmetries.

CP violation

The second important class of constraint provided by experimental information about known particles has to do with CP-violating effects, such as the electric dipole moments of the neutron and electron [15]. In Section 21.3 we saw that these effects are rather weak in the non-supersymmetric standard model with only one scalar doublet, apart from a potential problem with the parameter θ of quantum chromodynamics, discussed in

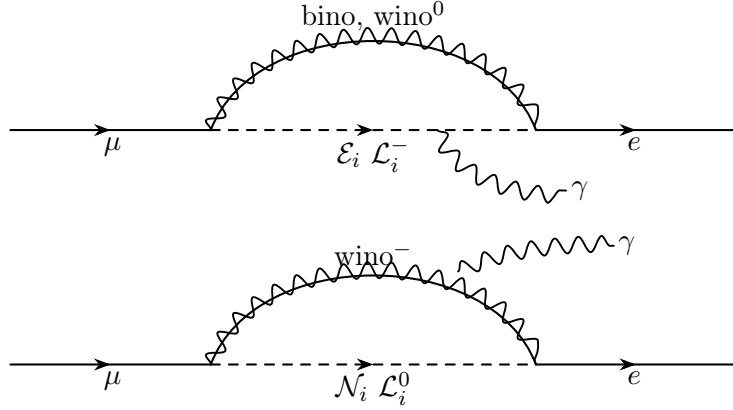


Figure 28.3. One-loop diagrams for the process $\mu \rightarrow e + \gamma$. Here solid lines are leptons; dashed lines are sleptons; combined solid and wavy lines are gauginos; and wavy lines are photons.

Section 23.6. This is because all CP-violating phases in the mass matrix of quarks and leptons and their interaction with gauge bosons could be absorbed into the definition of the quark and lepton fields if there were only two generations of quarks and leptons, and although there is a third generation, its mixing with the first two generations is (for mysterious reasons) quite weak. (This argument does not apply to processes that directly involve quarks of the third generation, such as B^0 - $\bar{B}^0$ mixing, to be measured in the planned ‘B factories.’) The electric dipole moment of the neutron in this simple non-supersymmetric version of the standard model is consequently expected [16] to be less than about $10^{-30} e \text{ cm}$, well below the experimental upper bound, $6.3 \cdot 10^{-26} e \text{ cm}$ [16a].

In contrast, the over 100 parameters of the minimum supersymmetric standard model in its most general form involve dozens of CP-violating relative phases. After integrating out the heavy superpartners of the known particles, these phases produce a number of CP-violating effective interactions to be added to the Lagrangian of the standard model. Those of minimum dimensionality, which on dimensional grounds are likely to be the most important, include electric dipole moments of the quarks and leptons [17], similar CP-violating ‘chromoelectric’ dipole moments

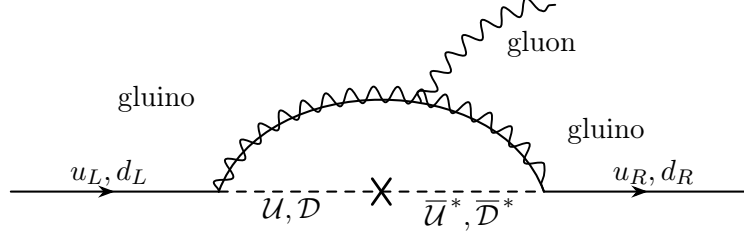


Figure 28.4. A one-loop diagram for the chromoelectric dipole moment of the u or d quarks. Here solid lines are quarks; dashed lines are squarks; combined solid and wavy lines are gluinos; and the wavy line is a gluon. The X represents the insertion of a bilinear interaction arising from a trilinear scalar field interaction combined with the spontaneous breakdown of $SU(2) \cdot U(1)$. There are also diagrams in which the gluon line is attached to one of the internal squark lines instead of to the gluino line.

contributing to the interactions of gluons with quarks [18], a CP-violating purely gluonic interaction [19], and a CP-violating interaction of the lightest Higgs scalar with leptons [20].

To take one example, consider the quark chromoelectric dipole moments, which in some models make the largest contribution to the electric dipole moment of the neutron. The CP-violating chromoelectric dipole moment operator is $(\bar{q}\gamma_5[\gamma_\mu, \gamma_\nu]\lambda_a q)f_a^{\mu\nu}$ (where q is a u or d color-triplet quark field, $f_a^{\mu\nu}$ is the $SU(3)$ field-strength tensor, and λ_a are the $3 \cdot 3$ generators of $SU(3)$). Since $\gamma_5[\gamma_\mu, \gamma_\nu]$ has matrix elements only between $\bar{q}_L$ and q_R or between $\bar{q}_R$ and q_L , in order for a one-loop graphs to contribute to the chromoelectric dipole moment an external left-handed u or d quark line must emit an internal gluino line and turn into a $\mathcal{U}$ or $\mathcal{D}$ squark line, which next turns into a $\bar{\mathcal{U}}^*$ or $\bar{\mathcal{D}}^*$ squark line, and then into a right-handed u or d quark line by absorbing the internal gluino line, with the external gluon line attached either to the internal gluino line or to one of the internal squark lines. (See Figure 28.4.)

To calculate this, we need to know the mixing of the scalar components $\mathcal{U}_i$ (or $\mathcal{D}_i$) of the left-chiral quark superfields Q_i with the complex conjugates $\bar{\mathcal{U}}_j^*$ (or $\bar{\mathcal{D}}_j^*$) of the scalar components of the left-chiral antiquark superfields $\bar{U}_j$ (or $\bar{D}_j$), produced by the spontaneous breakdown of $SU(2) \cdot U(1)$, and represented by the X in Figure 28.4. Part of this mixing arises from a contribution of the supersymmetric $\mathcal{F}$ -term interaction (28.1.7) to the last term in Eq. (26.4.7):

$$\mathcal{L}_{Q\bar{Q}\mathcal{H}} = - \left| \sum_{ij} h_{ij}^U \mathcal{U}_i \bar{\mathcal{U}}_j + \mu \mathcal{H}_1^0 \right|^2 - \left| \sum_{ij} h_{ij}^D \mathcal{D}_i \bar{\mathcal{D}}_j + \mu \mathcal{H}_2^0 \right|^2. \quad (28.4.7)$$

There is also a contribution from the A - and C -terms in Eq. (28.4.1):

$$\begin{aligned} \mathcal{L}'_{Q\bar{Q}\mathcal{H}} = & - \sum_{ij} h_{ij}^D \mathcal{D}_i \bar{\mathcal{D}}_j [-A_{ij}^D \mathcal{H}_1^0 + C_{ij}^D \mathcal{H}_2^{0*}] \\ & - \sum_{ij} h_{ij}^U \mathcal{U}_i \bar{\mathcal{U}}_j [A_{ij}^U \mathcal{H}_2^0 + C_{ij}^U \mathcal{H}_1^{0*}] - \text{H.c.} \end{aligned} \quad (28.4.8)$$

Replacing the neutral Higgs scalar fields with their expectation values gives the quadratic terms

$$\begin{aligned}\mathcal{L}_{\mathcal{Q}\overline{\mathcal{Q}}} = & -2 \operatorname{Re} \sum_{ij} m_{ij}^U \mathcal{U}_i \overline{\mathcal{U}}_j (\mu^* \cot \beta + A_{ij}^U + C_{ij}^U \cot \beta) \\ & -2 \operatorname{Re} \sum_{ij} m_{ij}^D \mathcal{D}_i \overline{\mathcal{D}}_j (\mu^* (\tan \beta)^* + A_{ij}^D - C_{ij}^D (\tan \beta)^*),\end{aligned}\quad (28.4.9)$$

where $m_{ij}^U = \langle \mathcal{H}_2^0 \rangle h_{ij}^U$ and $m_{ij}^D = -\langle \mathcal{H}_1^0 \rangle h_{ij}^D$ are the mass matrices of the quarks of charge $2e/3$ and $-e/3$, and

$$\tan \beta \equiv \frac{\langle \mathcal{H}_2^0 \rangle}{\langle \mathcal{H}_1^0 \rangle^*}. \quad (28.4.10)$$

Neglecting Cabibbo mixing, and for definiteness taking the A s and C s diagonal, Figure 28.4 makes contributions to the chromoelectric dipole moment of the u and d quarks of the form

$$d_u^{ce} = \frac{g_s^3}{16\pi^2} \operatorname{Im} [m_u A'_u I(m_{\mathcal{U}}, m_{\overline{\mathcal{U}}}, m_{\text{gluino}})], \quad (28.4.11)$$

$$d_d^{ce} = \frac{g_s^3}{16\pi^2} \operatorname{Im} [m_d A'_d I(m_{\mathcal{D}}, m_{\overline{\mathcal{D}}}, m_{\text{gluino}})], \quad (28.4.12)$$

where

$$A'_u \equiv (\mu^* + C_u) \cot \beta + A_u, \quad A'_d \equiv (\mu^* - C_d) (\tan \beta)^* + A_d, \quad (28.4.13)$$

and I is a complicated dimensionless function of its arguments arising from the integration over the virtual four-momentum. For $m_{\mathcal{Q}} \simeq m_{\overline{\mathcal{Q}}}$ and the gluino field defined to make m_{gluino} as well as $m_{\mathcal{Q}}$ real, the function I takes the form

$$I(m_{\mathcal{Q}}, m_{\mathcal{Q}}, m_{\text{gluino}}) = m_{\text{gluino}}^{-3} J \left(\frac{m_{\text{gluino}}^2}{m_{\mathcal{Q}}^2 - m_{\text{gluino}}^2} \right), \quad (28.4.14)$$

where [21]

$$J(z) = 2 \left(-z^4 + \frac{4}{3} z^3 + z^2 \right) \ln \left(\frac{1+z}{z} \right) + 2z^3 - \frac{11}{3} z^2. \quad (28.4.15)$$

The hard part of this sort of calculation always lies in estimating the contribution of an operator like the chromoelectric dipole interaction to hadronic matrix elements such as the electric dipole moment of the neutron. We can anticipate that there will be renormalization group corrections required because this operator is to be used at energies of the order of the neutron mass, rather than the masses of the squarks and gluino. More important is simply getting dimensional factors and factors of 4π right. For this purpose, it is usual to use a counting rule [22] known as ‘naive dimensional analysis.’ A connected graph with V_i vertices of type i and I internal lines will have a number L of loops given by $L = I - \sum_i V_i + 1$. If there are N_i lines attached to a vertex of type i and N external lines of the whole graph, then $2I + N = \sum_i V_i N_i$, so

$$L = 1 - \frac{N}{2} + \sum_i V_i \left(\frac{N_i}{2} - 1 \right).$$

We expect a factor of order $1/16\pi^2$ for each loop, so the coefficient of an operator $\mathcal{O}$ in the low-energy effective Lagrangian with N field factors will contain an over-all factor

$$(4\pi)^{N-2} \prod_i (4\pi)^{(2-N_i)V_i}.$$

If the operator $\mathcal{O}$ has dimensionality d and the interactions $\mathcal{O}_i$ of type i have dimensionality d_i , then the coefficient of $\mathcal{O}$ will have dimensionality $4 - d - \sum_i (4 - d_i)$, so this coefficient will also have a factor $M^{4-d} \prod_i M^{d_i-4}$, where M is some scale that is typical of hadronic physics, such as the nucleon mass or the energy $2\pi F_\pi \simeq 1200$ MeV, where the low-energy expansions discussed in Section 19.5 begin to break down. Finally, the contribution to the coefficient of $\mathcal{O}$ from some graph will of course be proportional to the couplings of all the operators $\mathcal{O}_i$ associated with the vertices in the graph. These remarks can be conveniently summarized by defining a ‘reduced coupling’: the reduced coupling associated with any operator $\mathcal{O}_i$ having N_i field factors, dimensionality d_i , and coupling constant g_i is

$$g_i^{\text{reduced}} \equiv g_i (4\pi)^{2-N_i} M^{d_i-4}. \quad (28.4.16)$$

The above estimates suggest the rule of naive dimensional analysis: the reduced coupling of any operator $\mathcal{O}$ in the effective hadronic Lagrangian is roughly equal to the product of the reduced couplings of the interactions that contribute to this effective coupling.

The neutron electric dipole moment is the coefficient of an operator with one photon and two neutron fields and dimensionality 5, so its reduced coupling is $M d_n^e / 4\pi$. Likewise, the quark chromoelectric magnetic moment has reduced coupling $M d_q^{ce} / 4\pi$. In addition to one factor of this reduced coupling, the reduced coupling of the neutron electric dipole operator must have a factor of the reduced coupling $e/4\pi$ of the electromagnetic coupling and an indeterminate number of factors of the reduced strong coupling $g_s/4\pi$, which at the low-energy scale M are not very different from unity and will be ignored. Taking the d quark contribution as representative of the contributions of both u and d quarks, the result then is that

$$d_n^e \approx \frac{e d_d^{ce}}{4\pi} \approx e \left(\frac{g_s}{4\pi} \right)^3 \text{Im} [m_d A'_d] I(m_{\mathcal{D}}, m_{\overline{\mathcal{D}}}, m_{\text{gluino}}). \quad (28.4.17)$$

Further simplifying by setting $m_{\text{gluino}} \simeq m_{\mathcal{D}} \simeq m_{\overline{\mathcal{D}}}$, so that $J = 7/18$, and taking $g_s^2/4\pi$ at the scale of squark and gluino masses to have the same value 0.12 as at m_Z , and $|m_d| \approx 7$ MeV, we have then

$$|d_n^e| \approx 0.5 \cdot 10^{-23} e \text{ cm} \frac{|A'_d| |\sin \varphi| \cdot (100 \text{ GeV})^2}{m_{\text{gluino}}^3}, \quad (28.4.18)$$

where φ is the phase of A'_d , with the convention that the gluino, quark, and squark masses are taken real. The contribution of the electric dipole moment of the quark is somewhat larger, while the contribution of the purely gluonic CP-odd operator is considerably smaller [23].

To avoid conflict with the experimental upper bound of $0.97 \cdot 10^{-25} e \text{ cm}$, either the phases associated with CP violation in the supersymmetric standard model must be less than about 10^{-2} , or some of the new particles of this model must be heavier than about

1 TeV. Similar conclusions have been reached from calculations of the electric dipole moments of atoms and molecules [23]. Even more stringent conditions on CP-violating phases have been derived [24] by considering the contribution of Figure 28.1 to the one precisely measured CP-violating effect, the imaginary part of the amplitude for K^0 - $\bar{K}^0$ oscillation.